Lecture 27 (*April 26*)

Reading: Section 7.1–7.2
Viewing: Lecture 29

We will now study the **Singular Value Decomposition (SVD)** of a matrix, which is very useful in computations. Think of image processing: in general, this deals with a large rectangular table of pixels, and the color of every pixel is an integer. So the image is encoded by an $m \times n$ matrix A of integers, where m and n are two very large numbers which keep track of how many pixels we have vertically and horizontally. If the colors of all the pixels are random, then there's not much you can do to simplify the matrix A. But in real life, nearby pixels tend to have similar colors, and patterns emerge that allow you to conclude that the matrix A has a special form. An extreme example is where your image consists only of three colors separated by vertical lines, such as a flag:

$$A = \begin{bmatrix} 1 & 1 & 2 & 2 & 2 & 2 & 2 & 3 & 3 & 3 \\ 1 & 1 & 2 & 2 & 2 & 2 & 2 & 3 & 3 & 3 \\ 1 & 1 & 2 & 2 & 2 & 2 & 2 & 3 & 3 & 3 \\ 1 & 1 & 2 & 2 & 2 & 2 & 2 & 3 & 3 & 3 \\ 1 & 1 & 2 & 2 & 2 & 2 & 2 & 3 & 3 & 3 \\ 1 & 1 & 2 & 2 & 2 & 2 & 2 & 3 & 3 & 3 \\ 1 & 1 & 2 & 2 & 2 & 2 & 2 & 3 & 3 & 3 \\ 1 & 1 & 2 & 2 & 2 & 2 & 2 & 3 & 3 & 3 \end{bmatrix}$$

Remembering the whole matrix is both memory and computational power-consuming, so you'd be more efficient by remembering instead the fact that A can be presented as the product:

$$A = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 & 2 & 2 & 2 & 2 & 2 & 3 & 3 & 3 \end{bmatrix}$$

of a column vector and a row vector. In general, the goal of SVD is to write any rectangular matrix as a sum of such products (let r be the rank of A):

$$\boxed{A = \boldsymbol{u}_1 \sigma_1 \boldsymbol{v}_1^T + \cdots + \boldsymbol{u}_r \sigma_r \boldsymbol{v}_r^T} \tag{241}$$

where each $\boldsymbol{u}_i$ is a column vector, each $\boldsymbol{v}_i^T$ is a row vector, and:

the positive numbers $\boxed{\sigma_1, \ldots, \sigma_r > 0}$ **are called singular values**

Let us now explain how to obtain the decomposition (241).

Fact 22. *The $m \times m$ matrix AA^T and the $n \times n$ matrix A^TA are both symmetric and positive semidefinite. These two matrices have the same set of positive eigenvalues; call them $\sigma_1^2, \ldots, \sigma_r^2$.*

Proof: assume that $\lambda \neq 0$ is an eigenvalue of AA^T. Then:

$$AA^T\boldsymbol{v} = \lambda\boldsymbol{v} \qquad \Rightarrow \qquad A^TA(A^T\boldsymbol{v}) = \lambda(A^T\boldsymbol{v})$$

which implies that λ is also an eigenvalue of A^TA corresponding to the eigenvector $A^T\boldsymbol{v}$ (the reason you need $\lambda \neq 0$ is in order to infer that $A^T\boldsymbol{v} \neq 0$, since eigenvectors must be non-zero by definition).

Since symmetric matrices (such as AA^T and A^TA) have orthonormal eigenvectors, we can choose:

$$\begin{aligned} &\textbf{left singular vectors } \ \boldsymbol{u}_1, \ldots, \boldsymbol{u}_r, \ldots, \boldsymbol{u}_m \in \mathbb{R}^m \\ &\textbf{right singular vectors } \boldsymbol{v}_1, \ldots, \boldsymbol{v}_r, \ldots, \boldsymbol{v}_n \in \mathbb{R}^n \end{aligned}$$

such that the $\boldsymbol{u}_i$'s are orthonormal and the $\boldsymbol{v}_i$'s are orthonormal, and:

$$AA^T\boldsymbol{u}_i = \sigma_i^2\boldsymbol{u}_i \qquad \text{and} \qquad A^TA\boldsymbol{v}_i = \sigma_i^2\boldsymbol{v}_i \tag{242}$$

(the formulas above even hold for $i > r$, if we let $\sigma_{r+1}, \sigma_{r+2}, \cdots = 0$). But we actually want the left and right singular vectors to also be connected, via the following formulas:

$$\boxed{A\boldsymbol{v}_i = \sigma_i\boldsymbol{u}_i} \qquad \text{and} \qquad \boxed{A^T\boldsymbol{u}_i = \sigma_i\boldsymbol{v}_i} \tag{243}$$

which also hold for all i, even $i > r$. The numbers $\sigma_1, \ldots, \sigma_r$ are called **singular values**. It is easy to see that (243) implies both (242) and (241): the former implication is an easy exercise, while the latter follows by multiplying both sides of (241) with any given vector $\boldsymbol{v}_i$ and using that $\boldsymbol{v}_j^T\boldsymbol{v}_i = 0$ if $j \neq i$. Note that the singular value decomposition (241) can be written in compact form as:

$$\boxed{A = U\Sigma V^T} \tag{244}$$

where:

$$U = \left[\ \boldsymbol{u}_1 \mid \ldots \mid \boldsymbol{u}_m\ \right] \qquad \Sigma = \begin{bmatrix} \sigma_1 & 0 & 0 & 0 & \ldots \\ 0 & \ddots & 0 & 0 & \ldots \\ 0 & 0 & \sigma_r & 0 & \ldots \\ 0 & 0 & 0 & 0 & \ldots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix} \qquad V = \left[\ \boldsymbol{v}_1 \mid \ldots \mid \boldsymbol{v}_n\ \right]$$

Remark. *From* (243)*, it follows that* $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r$ *form a basis of the column space of* A*, and* $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_r$ *form a basis of the row space. The orthogonality of the four subspaces implies that* $\boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_m$ *form a basis of the left nullspace of* A*, while* $\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_n$ *form a basis of the nullspace.*

Since V is an orthogonal matrix, we have $V^T = V^{-1}$, so (244) implies:

$$\boxed{AV = U\Sigma} \tag{245}$$

which is just another way to write the first identity in (243). As for the second identity in (243), we can obtain it as follows. First transpose relation (244):

$$A^T = V\Sigma^TU^T \tag{246}$$

Since U is an orthogonal matrix, we have $U^T = U^{-1}$, hence:

$$A^T U = V \Sigma^T \tag{247}$$

Therefore, the SVD of the transpose of A is obtained by transposing the SVD of A.

Let's do an example of SVD, which will also show us that the procedure works, for the matrix:

$$A = \begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix}$$

We first need to compute the singular values, which will be the square roots of the eigenvalues of:

$$A^T A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix} = \begin{bmatrix} 17 & 8 \\ 8 & 17 \end{bmatrix}$$

The characteristic polynomial of this matrix is:

$$p(\lambda) = \det \begin{bmatrix} 17-\lambda & 8 \\ 8 & 17-\lambda \end{bmatrix} = (17-\lambda)^2 - 8^2$$

and its roots will be precisely those numbers for which $17 - \lambda = \pm 8$, so:

$$\begin{aligned} 17 - \sigma_1^2 = -8 \quad &\Rightarrow \quad \sigma_1^2 = 25 \quad \Rightarrow \quad \sigma_1 = 5 \\ 17 - \sigma_2^2 = 8 \quad &\Rightarrow \quad \sigma_2^2 = 9 \quad \Rightarrow \quad \sigma_2 = 3 \end{aligned}$$

(it is customary to order your singular values $\sigma_1, \ldots, \sigma_r$ from the greatest to the smallest). Next, you need to find the right singular vectors, which will be eigenvectors of the matrix $A^T A$:

$$\begin{aligned} \boldsymbol{v}_1 &\in N\left(\begin{bmatrix} 17-\sigma_1^2 & 8 \\ 8 & 17-\sigma_1^2 \end{bmatrix}\right) = N\left(\begin{bmatrix} -8 & 8 \\ 8 & -8 \end{bmatrix}\right) \quad \Rightarrow \quad \boldsymbol{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ \boldsymbol{v}_2 &\in N\left(\begin{bmatrix} 17-\sigma_2^2 & 8 \\ 8 & 17-\sigma_2^2 \end{bmatrix}\right) = N\left(\begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix}\right) \quad \Rightarrow \quad \boldsymbol{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \end{aligned}$$

Note that $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ are orthonormal (indeed, recall that eigenvectors of symmetric matrices corresponding to different eigenvalues are always orthogonal, and we have intentionally scaled them so as to have length 1). Now let's compute the left singular vectors. Formula (243) forces us to take:

$$\begin{aligned} \boldsymbol{u}_1 &= \frac{1}{5} A \boldsymbol{v}_1 = \frac{1}{5} \begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \\ \boldsymbol{u}_2 &= \frac{1}{3} A \boldsymbol{v}_2 = \frac{1}{3} \begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{3\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix} \end{aligned}$$

The vectors $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$ are also orthonormal (this follows from the fact that $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ are) but we need to complete them to an orthonormal basis of $\mathbb{R}^3$. This is achieved by Gram-Schmidt. Start with any vector that is not a linear combination of $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$, say:

$$\boldsymbol{a}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

and construct the vector:

$$\boldsymbol{w}_3 = \boldsymbol{a}_3 - \text{proj}_{\boldsymbol{u}_1}\boldsymbol{a}_3 - \text{proj}_{\boldsymbol{u}_2}\boldsymbol{a}_3 = \begin{bmatrix}0\\0\\1\end{bmatrix} - \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\\0\end{bmatrix}\cdot 0 - \frac{1}{3\sqrt{2}}\begin{bmatrix}1\\-1\\4\end{bmatrix}\cdot\frac{4}{3\sqrt{2}} = \frac{1}{9}\begin{bmatrix}-2\\2\\1\end{bmatrix}$$

The vector $\boldsymbol{w}_3$ thus constructed is orthogonal to both $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$, so we need to renormalize it:

$$\boldsymbol{u}_3 = \frac{\boldsymbol{w}_3}{||\boldsymbol{w}_3||} = \frac{1}{3}\begin{bmatrix}-2\\2\\1\end{bmatrix}$$

With this in mind, we conclude the following SVD for the matrix A:

$$A = \begin{bmatrix}\frac{1}{\sqrt{2}} & \frac{1}{3\sqrt{2}} & -\frac{2}{3}\\ \frac{1}{\sqrt{2}} & -\frac{1}{3\sqrt{2}} & \frac{2}{3}\\ 0 & \frac{4}{3\sqrt{2}} & \frac{1}{3}\end{bmatrix}\begin{bmatrix}5 & 0\\0 & 3\\0 & 0\end{bmatrix}\begin{bmatrix}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}}\end{bmatrix}$$

So in general, what can we say about a rank 1 matrix? The answer is that it will be of the form:

$$A = \boldsymbol{u}\sigma\boldsymbol{v}^T \tag{248}$$

where $\boldsymbol{u}$ and $\boldsymbol{v}$ are length 1 vectors, and σ is a non-zero number. In other words, any column of a rank 1 matrix is a multiple of $\boldsymbol{u}$ and any row is a multiple of $\boldsymbol{v}$. Obviously, you can rescale $\boldsymbol{u}$ and $\boldsymbol{v}$ such that they have length 1 (at the cost of changing the number σ) and you can flip the sign on σ (at the cost of replacing $\boldsymbol{v}$ by $-\boldsymbol{v}$) so as to ensure $\sigma > 0$. With this in mind, the singular value decomposition (241) can be thought of as:

SVD writes any matrix as a sum of rank 1 matrices (249)

where the columns (respectively rows) of the rank 1 matrices involved are orthonormal. This is good to keep in mind, since it can save you a lot of time when doing SVD on a matrix where it's intuitively pretty obvious how to break it up as a sum of rank 1 matrices. For example:

$$A = \begin{bmatrix}0 & 5 & 0\\-3 & 0 & 0\end{bmatrix}$$

If you started doing the usual algorithm (compute the eigenvalues and eigenvectors of the 3×3 matrix A^TA) you might find yourself doing a lot of work. You can save a lot of time by observing that A decomposes as:

$$A = \begin{bmatrix}0 & 5 & 0\\0 & 0 & 0\end{bmatrix} + \begin{bmatrix}0 & 0 & 0\\-3 & 0 & 0\end{bmatrix}$$

Each of the two matrices above has rank 1, because all of their columns are multiples of a single vector. Even more explicitly, we have:

$$A = \underbrace{\begin{bmatrix}1\\0\end{bmatrix}}_{\boldsymbol{u}_1}\cdot\underbrace{5}_{\sigma_1}\cdot\underbrace{\begin{bmatrix}0 & 1 & 0\end{bmatrix}}_{\boldsymbol{v}_1} + \underbrace{\begin{bmatrix}0\\-1\end{bmatrix}}_{\boldsymbol{u}_2}\cdot\underbrace{3}_{\sigma_2}\cdot\underbrace{\begin{bmatrix}1 & 0 & 0\end{bmatrix}}_{\boldsymbol{v}_2}$$

However, when using this shortcut, you need to make sure that the vectors $\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots$ you construct are orthonormal (and also that the vectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots$ are orthonormal).